

Dot Product	Cross Product of Conformable Matrices
Matrix-Vector Multiplication	Matrix-Matrix Multiplication
Range/Column Space	Nullspace
Rank	Theorem 1.2
Nonsingular/Invertible	A Matrix Inverse Times a Vector

$\mathbf{A}^\top \mathbf{B}$	$\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$
For the matrix-matrix product $B = AC$, each column of B is a linear combination of the columns of A. Let A be $l \times m$, C be $m \times n$, then B is $l \times n$ with entries defined by $b_{ij} = \sum_{k=1}^m a_{ik} c_{kj}$ $\mathbf{b}_j = \mathbf{A} \mathbf{c}_j = \sum_{k=1}^m c_{kj} \mathbf{a}_k$	Let x be an n-dimensional column vector and let A be an $m \times n$ matrix. Then the matrix-vector product $b = Ax$ is the m-dimensional column vector defined as follows: $b_i = \sum_{j=1}^n a_{ij} x_j, \quad i = 1, \dots, m$ $\mathbf{b} = \mathbf{A} \mathbf{x} = \sum_{j=1}^n x_j \mathbf{a}_j$
The nullspace of $\mathbf{A} \in \mathbb{C}^{m \times n}$, written $\text{null}(\mathbf{A})$, is the set of vectors $\mathbf{x}$ that satisfy $\mathbf{A} \mathbf{x} = \mathbf{0}$, where $\mathbf{0}$ is the 0-vector in $\mathbb{C}^m$	• The range of a matrix $\mathbf{A}$, written $\text{range}(\mathbf{A})$, is the set of vectors that can be expressed as $\mathbf{A} \mathbf{x}$ for some $\mathbf{x}$. • The space spanned by the columns of $\mathbf{A}$
A matrix $\mathbf{A} \in \mathbb{C}^{m \times n}$ with $m \geq n$ has full rank if and only if it maps no two distinct vectors to the same vector	Let $\mathbf{A}$ be an $m \times n$ matrix such that $m \geq n$. Then $\mathbf{A}$ has n linearly independent columns with rank n
• $\mathbf{A}^{-1} \mathbf{b}$ is the vector of coefficients of the expansion of $\mathbf{b}$ in the basis of columns of $\mathbf{A}$ • NOTE: Multiplication by $\mathbf{A}^{-1}$ is a change of basis operation	A matrix is invertible if it is a square matrix of full rank